Tokenizing Real-World Assets

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CBER CTE Conference

October 30, 2026

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Programmability and continuous tradability do not guarantee liquidity. Buyers, market depth, and economic function matter.

The Rise of Tokenized Real-World Assets

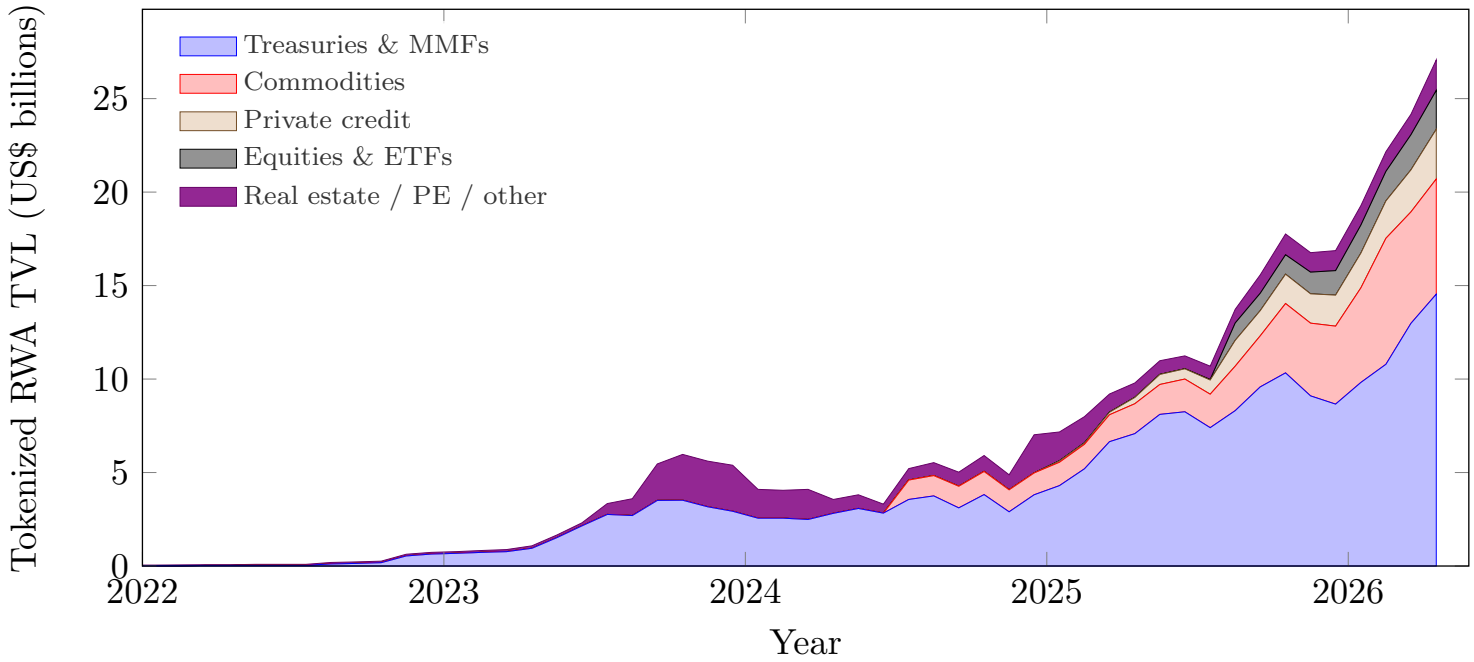


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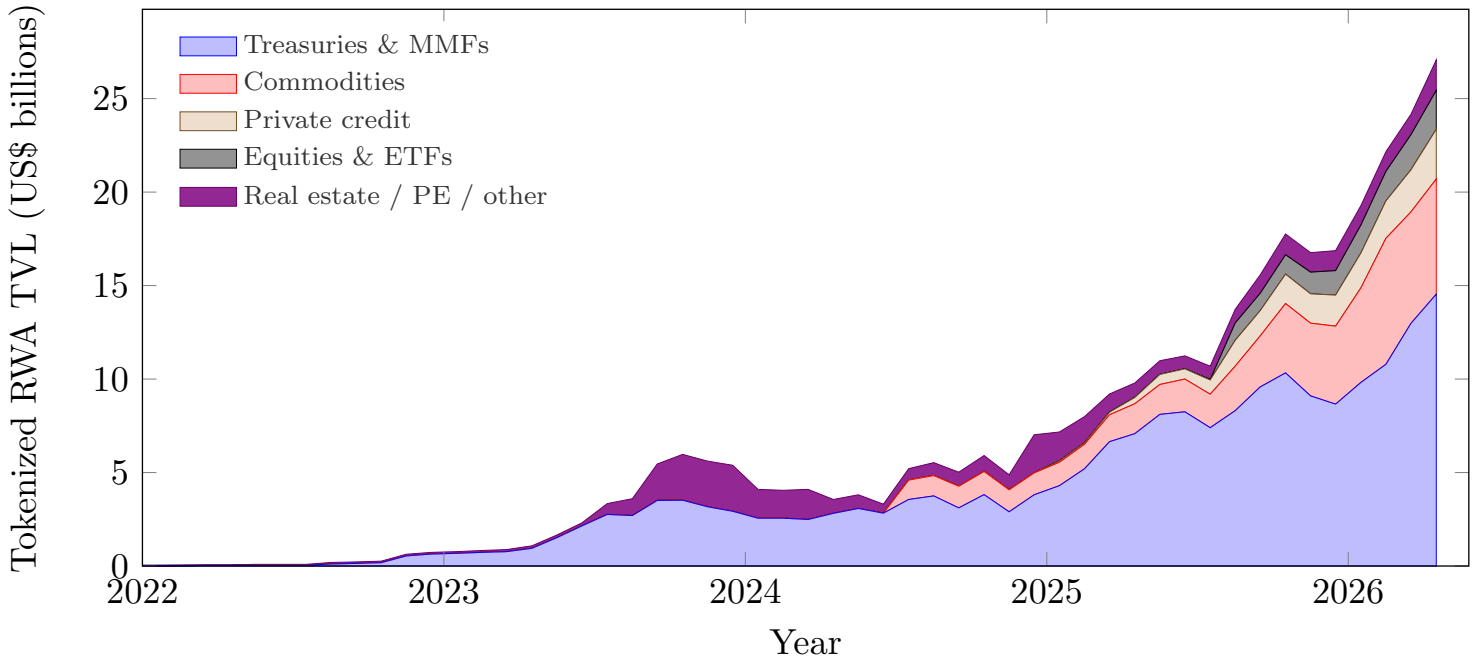


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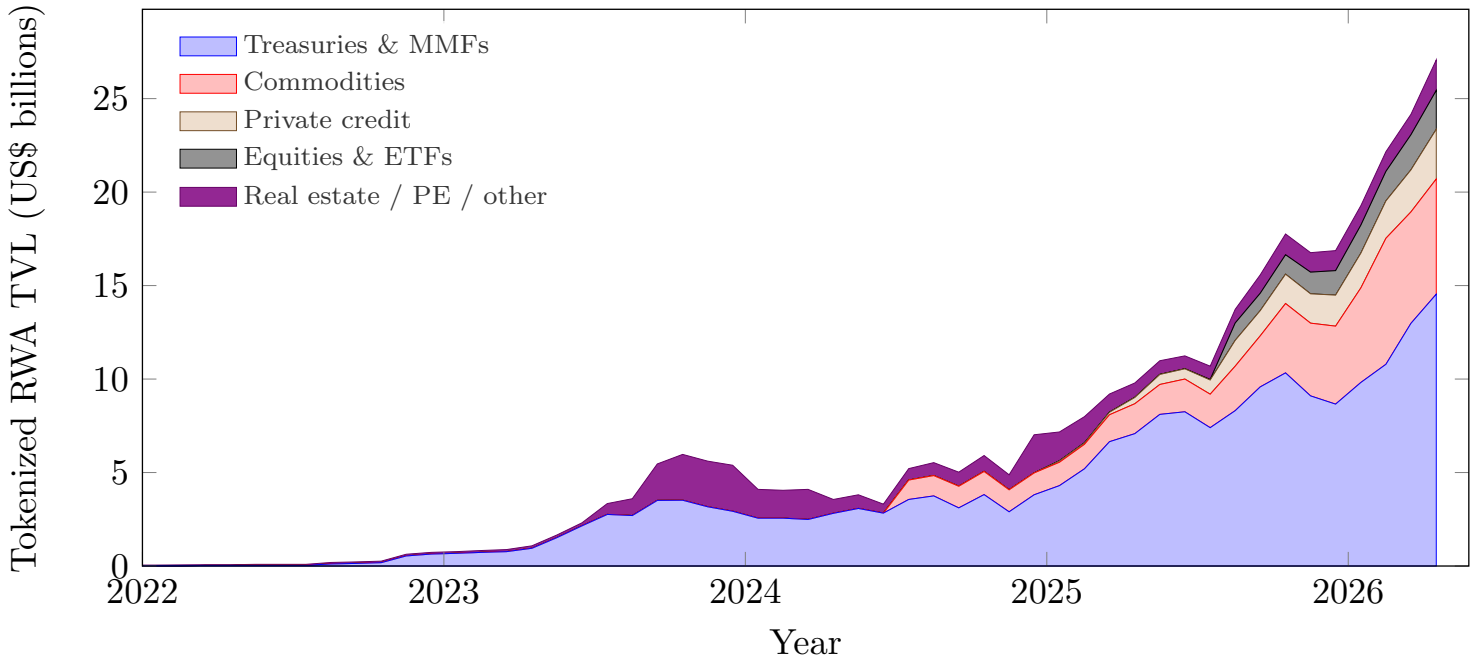


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Concentrated issuance: The top ten issuers account for about 82% of total value including Figure. Figure alone accounts for about 37%.

Three Economic Categories

Category	**Examples**	**Economic function**
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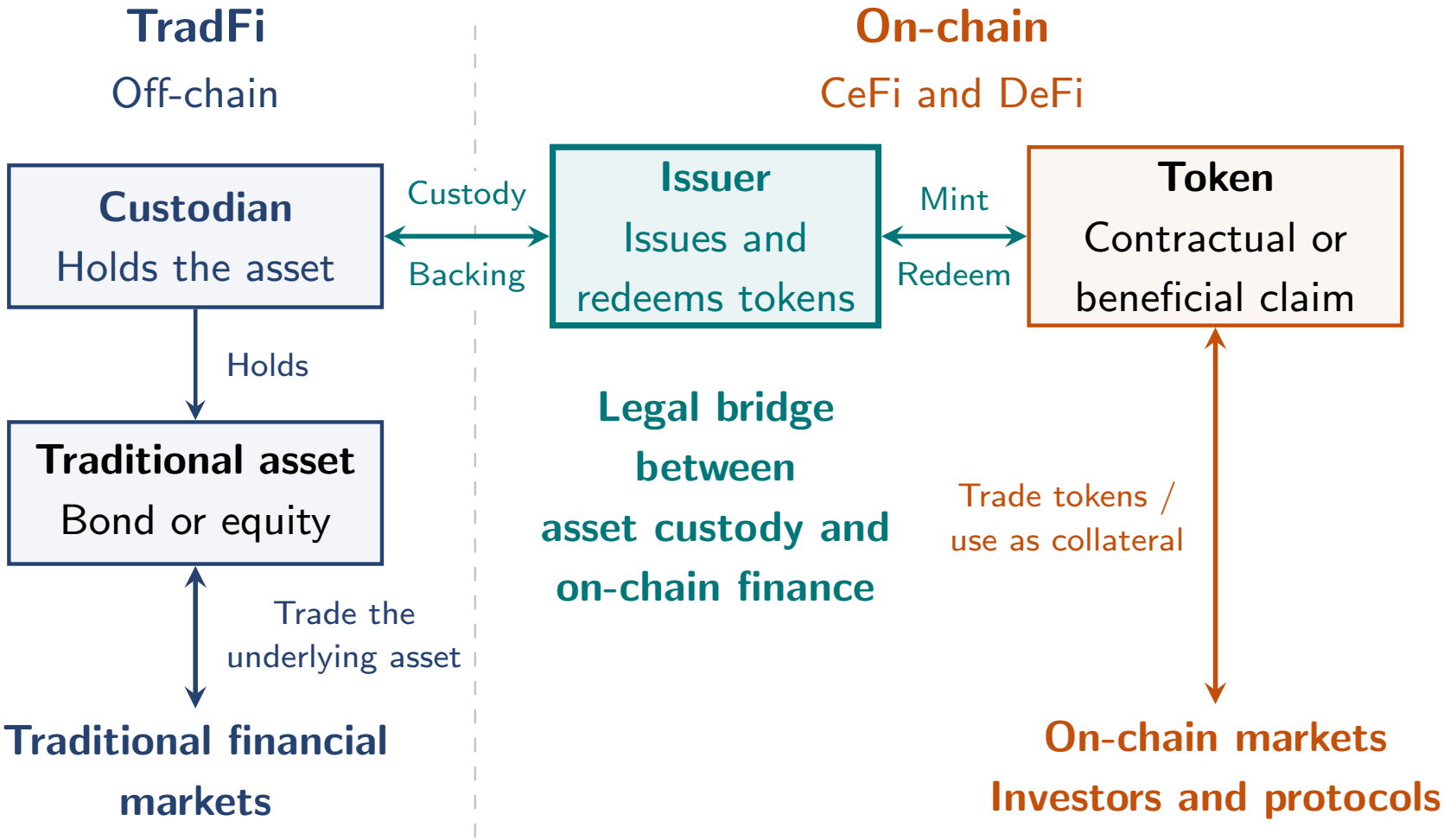
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Economic function and underlying liquidity determine the relevant risks.

The Issuer Links Custody and On-Chain Markets



Legal structure defines claim rights, asset backing, and redemption terms.

The Token Holder's Legal Claim and Eligibility

The token holder's claim

Contractual claims, beneficial interests, and direct title differ. Law, authoritative records, and transaction documents define enforceable rights.

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Tokenization changes trading, programmability, and settlement. Underlying risks and cash flows remain.

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DeFi depends on TradFi: automated execution still depends on reliable information and accountable institutions in the traditional financial system.

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Demand segment	**Economic use**	**What matters for design**
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Investor composition matters for both risks and benefits.

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Asset and scale	**On-chain use**	**What supports price alignment**
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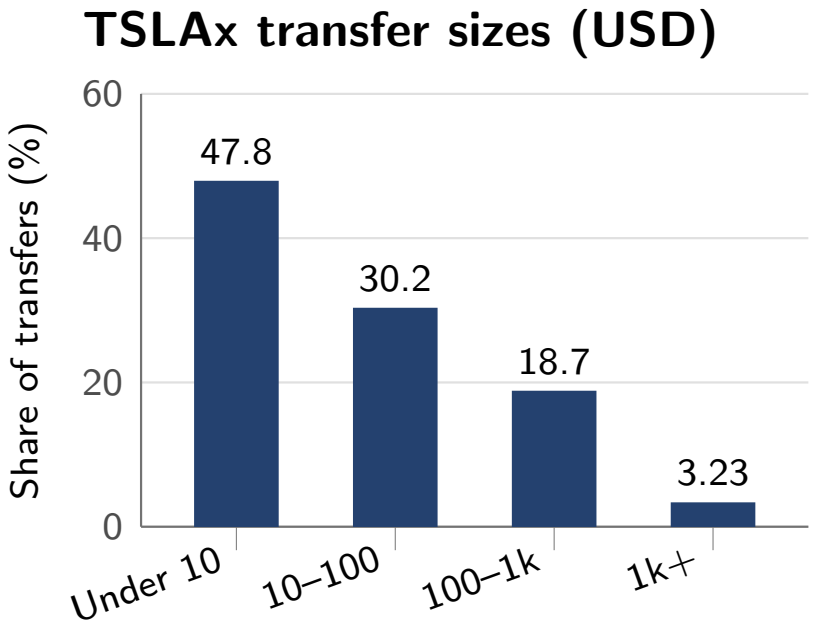
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Integration is the primary use. Effects on underlying-market liquidity remain an open question. Arbitrage mechanism with traditional markets keeps prices aligned.

Liquid Assets: Participation and Pricing Evidence

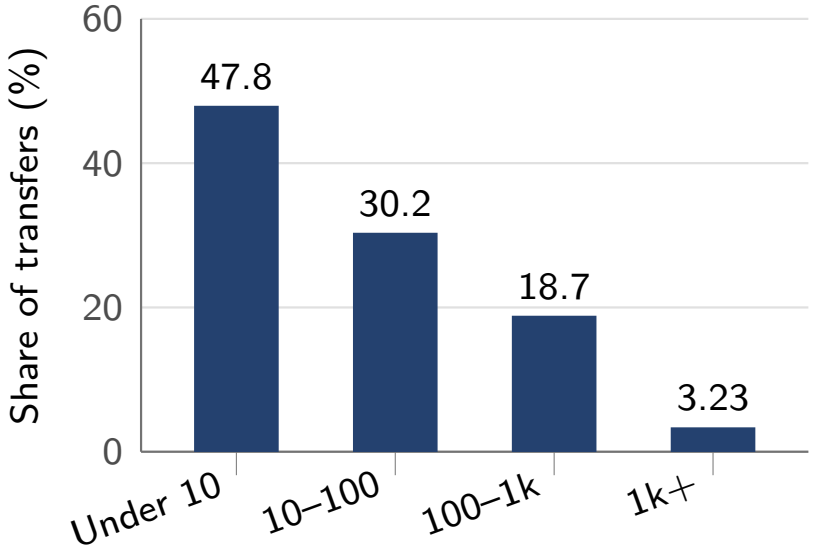


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TSLAx transfer sizes (USD)



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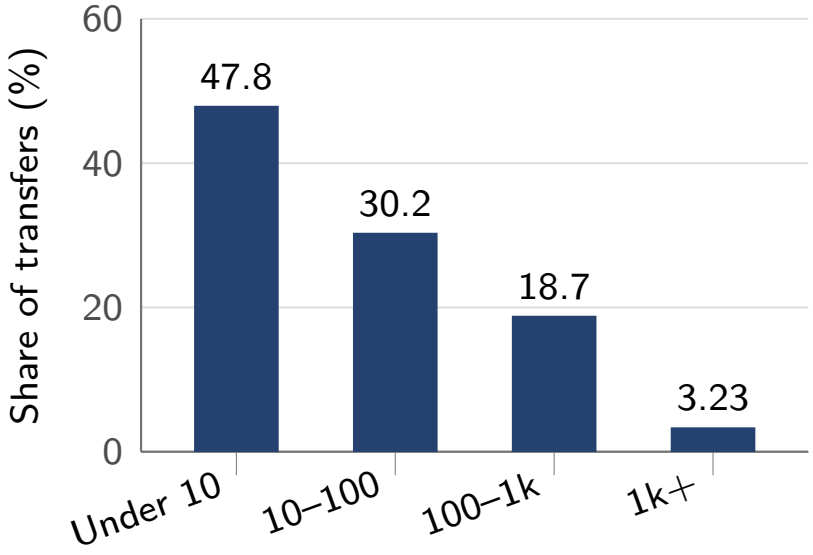
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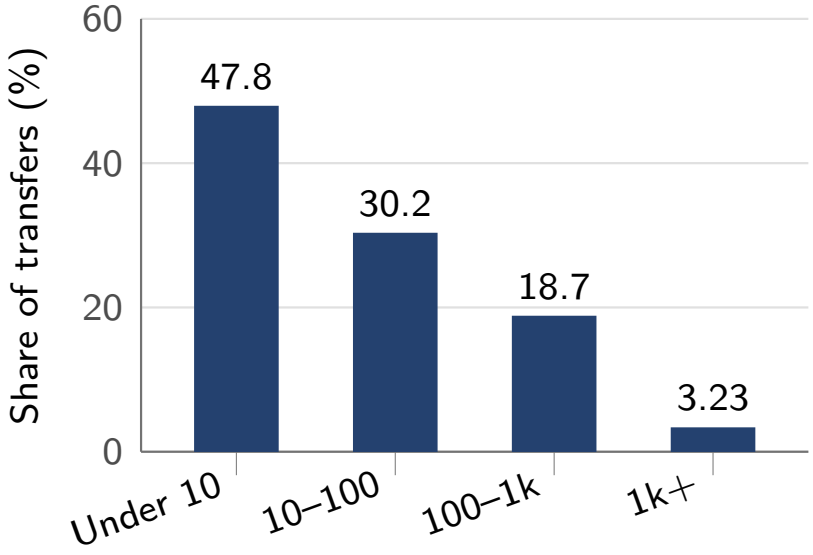
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Token prices closely track the underlying. Liquidity varies across tokens and venues.

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Tokenized Treasury/MMF fund are also money-like

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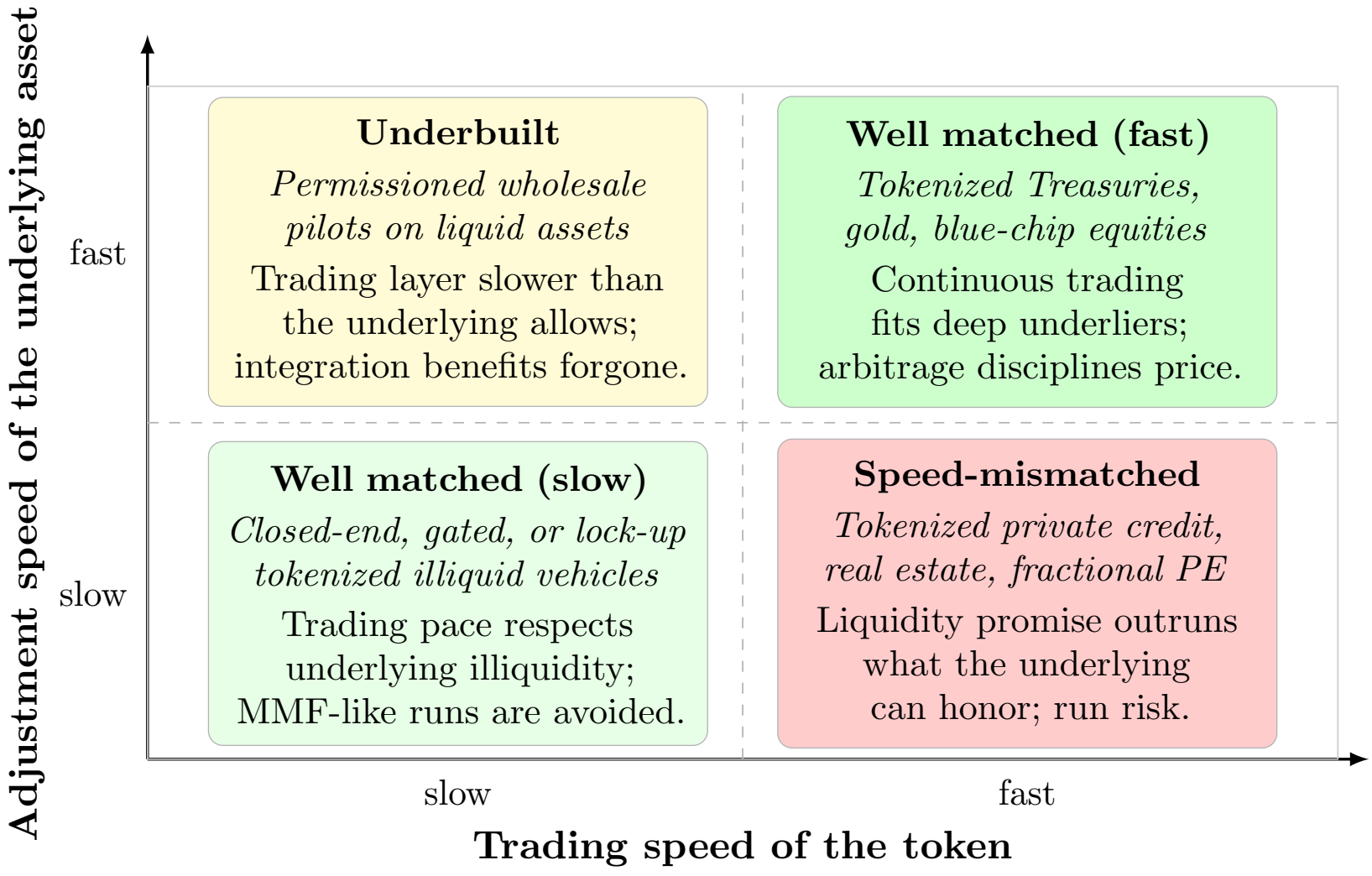
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Tokenized Structures: ETF-like tradability can coexist with mutual fund-like redemption constraints.

Policy Proposal: The Speed-Matching Principle



Speed Matching across Asset Categories

Redemption commitments must fit funding capacity. Trading and collateral rules must reflect valuation and market depth.

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Closed-end claims can trade continuously without promising issuer redemption. Redemption often not feasible.

Implementing Speed Matching

Proposed baseline standards for policy calibration

Redemption and funding capacity

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Calibration depends on the asset, claim design, and collateral use.

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Stress transmission depends on leverage and contractual links to the underlying balance sheet.

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Key empirical question: what are the benefits of tokenization, and the implementation difficulties in practice? What is it used for?

Future Research Questions

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Comparable data on holders, venues, and redemptions can test the proposed mechanisms.

Minimum Design Standards: Selected Details

Dimension	**Illustrative design standards**
Legal claim and custody	Identify the asset, issuer, servicer, claim rights, and bankruptcy segregation.
Redemption and pricing	Liquid tokens: qualified-holder windows. Private credit: notice/gates. Private alternatives: closed-end, interval or auction designs.
Valuation and oracles	Daily NAV for Treasury/MMF tokens. Private-credit NAV at least monthly. Stronger independent price inputs for collateral use.
Venues and interoperability	Report execution and volume quality, depth, spreads, issuer support, provider concentration, and cross-chain fungibility.
Eligibility and protocol use	Specify holder restrictions, checks, and approvals while permitting eligible smart contracts to use the token.

Pilots and Interoperability: Evidence to Collect

International initiatives include the UK Digital Securities Sandbox, EU DLT Pilot, Singapore's Project Guardian, and Switzerland's DLT framework.

Market outcomes as well as operational feasibility

Report investor composition, venue-level liquidity, redemption use, oracle updates, and cross-chain issuance.

Rights and settlement must travel together

Common token interfaces help only when custody, redemption, legal rights, and finality remain clear across chains and jurisdictions.

Pilot scope and redemption design

Early pilots should prioritize liquid and money-like assets. The paper finds redemption design largely left to issuers, leaving a calibration gap.

Comparable reporting can turn pilots into evidence for asset-specific policy.

Tokenization versus ETFs: Six Design Differences

	ETF benchmark	Tokenized claim
Contract use	Conventional custody and trading	Automated distributions, compliance, and collateral use
Arbitrage	Authorized participants	Qualified holders, code, or issuer discretion
Trading pairs	Conventional venue arrangements	Securities, stablecoins, and crypto in one protocol
Illiquid assets	Managed baskets	Private claims with limited in-kind exit
Cross-chain use	No native chain fragmentation	Versions may lack cross-chain fungibility
Price inputs	Market prices and fund NAV	Oracles can govern collateral and redemption

Atomic settlement and shared records can lower operating costs.

PORTS: A Safe-Asset Proposal

Proposed instrument

Perpetual Overnight Rate Treasury Securities would combine overnight floating-rate returns with daily par issuance and redemption.

Potential role

A transparent reserve or collateral asset for stablecoins and tokenized cash-management instruments.

Issuer safeguards remain necessary

A better backing asset still requires credible custody, redemption terms, and controls on collateral use.

PORTS is a proposal for safe-asset design, separate from the regulation of token issuers.

Wholesale versus Retail Tokenization

	Wholesale	**Retail-facing**
Participants	Bank treasuries, dealers, institutions	Individual investors and wallets
Typical setting	Permissioned institutional networks	Public-chain distribution and trading
Main uses	Interbank settlement, repo, collateral mobility	Investment access and fractional exposure
Policy emphasis	Finality, resilience, governance, and access	Claim rights, custody, suitability, and venue integrity

The same underlying asset can offer different benefits and require different safeguards across institutional and retail settings.

Speed Matching and Liquidity Promises

Claim design	Secondary trading	Issuer redemption
Liquid-asset token	May trade continuously, including off-hours	Product-specific windows and eligible holders
Closed-end private-asset claim	Can trade continuously at market prices	No routine redemption. Exit depends on finding buyers.
Private-asset fund with redemptions	May trade at market prices between redemption windows	Notice, gates, or periodic windows should reflect valuation and sale capacity.

Valuation has its own clock: Collateral rules must account for stale NAVs and uncertain prices.

Redemption commitments must fit funding capacity. Secondary trading can clear at market prices.

Illiquid Assets: Secondary-Market Design

Mechanism	Service it provides	Economic constraint
Automated market maker	Immediate execution, often for small transactions	Pool depth and compensated liquidity providers
Posted offers / P2P	Negotiated or posted trades, including larger transfers	Finding buyers and observing available depth
Issuer buyback	A price or exit backstop within stated limits	Funding capacity, caps, and appraisal-based pricing

In a tokenized real-estate platform, small trades use automated pools while P2P accounts for most dollar volume.

More venues can improve execution while fragmenting thin liquidity. Execution and volume quality need scrutiny.

Policy Analogy: Stablecoins and Private Claims

The paper's GENIUS Act analogy

Licensed issuers, liquid reserves, attestations, and par redemption support payment stablecoins. Licensing also limits decentralized designs.

Yield-bearing claims

Extending the interest prohibition to Treasury/MMF tokens would remove their yield use case.

Private claims need fund-style safeguards

Closed-end or interval designs, valuation controls, and exit-cost allocation can align redemption with slow-moving assets.

Speed matching requires different safeguards for payments, liquid funds, and private claims.